

Leeward — Veteran Care Continuity Proposal

2026-09-19 · @Someone

Summary

Leeward turns a climate forecast into a ranked *action* list for VA care teams: who to call, when, and why, given how many calls the team can actually make. The base proposal predicts risk. Leeward predicts risk *and* what a call changes, then allocates scarce outreach where it averts the most harm.

The pitch in one line: *"Care that gets ahead of the weather."* Every veteran already has a care team; Leeward tells that team which 40 of its 1,200 patients to reach before Thursday's heat wave, and hands each one a plan that is verifiably from the VA.

What changed from the base proposal, and why:

Base proposal	Leeward	Why judges care
Weekly logistic risk score per veteran	Daily discrete-time hazard with 0–3 day heat/smoke lags and a decision layer that ranks <i>actions</i> under staff capacity	Heat kills over 1–3 days, not a week; a list of 10,000 risks is not a plan
Cumulative exposure as a hand-built sum	Latent exposure dose with measurement error, so uncertain burn-pit history propagates into predictions	Honest about the biggest data gap; shows why Bayesian
Covers opportunity areas 01–03	Adds 04 (coordination export to partners) and 05 (verified outreach, anti-scam card)	Challenge text names 5 areas; most teams will do 3
Uncertainty shown as a range	Uncertainty <i>drives</i> triage: high-uncertainty veterans get a cheap check-in call (value of information)	Turns the Bayesian story from decoration into a feature

Base proposal	Leeward	Why judges care
Demo = map + list	Demo = one veteran's story, then the map, then the care-team screen, then the fairness audit	Impact is worth 5 of 22 rubric points; story carries impact

Everything runs on public data and a synthetic NYC cohort of 10,000 veterans. No real patient data is used.

What judges will poke at in the base proposal

The base proposal is strong on story (WTC Registry as blueprint) and weak on the two things the challenge text stresses: *coordination* and *what happens next*. Last year's rubric weighted Impact 5, AI/Data 4, Team 4, Innovation 3, Feasibility 3, Scalability 3 ([2025 Devpost rubric](#)); assume similar weights.

Weak spot	The likely judge question	Leeward's answer
"Another risk score"	VA already has the CAN score and HHS emPOWER already counts oxygen-dependent people by ZIP. What is new?	The climate × exposure <i>interaction</i> and the capacity-aware action list. Leeward consumes emPOWER, it does not rebuild it
Weekly resolution	A heat wave lasts 3 days and deaths lag exposure by 0–3 days. A weekly score cannot time a Tuesday call	Daily hazard model with distributed lags; alerts fire 2–5 days ahead off NWS forecasts
Only areas 01–03	The brief lists five gaps and says coordination and exploitation are the unmet needs	Partner export (04) and verified outreach + scam shield (05) are in the demo, not the appendix
10,000-row ranked list	Which care team can act on 10,000 rows?	Ranking is over <i>actions</i> , cut at the team's daily capacity, with an explicit harm-averted number per call

Weak spot	The likely judge question	Leeward's answer
WTC priors as fact	"You are assuming 9/11 survivors and veterans behave alike"	Priors are weakly informative and the demo shows a prior-sensitivity panel; the model recovers the truth from synthetic data either way
No loop closure	Did the call happen? Did it help?	Outcome of each action is logged and feeds the next posterior update
Synthea cohort is national	Synthea places patients by US county, not NYC ZIPs	Cohort is re-homed to NYC ZIPs weighted by ACS veteran counts; documented in the generator
MCMC on $10k \times \text{days} \times 4 \text{ outcomes}$	Will it finish before Sunday?	Fit on binomial cell counts ($\text{ZIP} \times \text{stratum} \times \text{day}$), NumPyro on JAX, posterior cached; live demo scores in milliseconds

One more risk: the base proposal cites two news pieces as if they were registry publications. Cite the NYC Health Department and PACT Act pages directly; judges from Cornell will check.

The reframe: from risk forecast to continuity-of-care decision engine

Leeward answers three questions a care team asks before a climate event, in order: **who** is at risk, **when** the risk peaks, and **what changes if we act**. The base proposal stops after the first.

flowchart LR

```

A[NWS + AirNow<br/>3-7 day forecast] --> B[Hazard model<br/>daily risk per veteran]
C[Synthetic cohort<br/>health + exposure] --> B
D[emPOWER, HVI, SVI<br/>ZIP context] --> B
B --> E[Decision layer<br/>harm averted per action]
F[Team capacity<br/>calls, refills, rides] --> E
E --> G[Today's action list<br/>cut at capacity]
G --> H[Verified outreach<br/>VA channels + scam shield]
H --> I[Outcome log] --> B

```

Forecast in, capacity in, a short verified action list out; every logged outcome updates the model.

The decision layer. For each veteran i , action a and day t , the model computes expected harm averted:

$$\mathrm{EHA}_{i,a,t} = \sum_k w_k \; ; \; \mathbb{E}_{\{\text{posterior}\}} \left[h_{ikt} \cdot \tau_{ak} \right]$$

where h is the posterior daily hazard for need k , w_k is a severity weight set with clinicians (a treatment gap for a chemo patient outweighs a missed wellness text), and τ_{ak} is the fraction of need k that action a prevents. τ is a prior from published emergency-outreach effects, stated openly and adjustable in the UI. Actions are then chosen greedily under the team's capacity (calls per day, refills, rides), which is a knapsack the team can re-run when staffing changes.

Uncertainty as a feature. The posterior splits into two parts: risk we are sure about and risk we are unsure about. A veteran with a wide interval (unknown AC status, unknown deployment history) gets a cheap information action, a 3-minute check-in call, because the expected value of that information is high. A veteran with a narrow, high interval gets the expensive action, an early refill and a cooling-center ride. Most teams will show a range; Leeward acts on it.

Tiers the demo shows.

Tier	Trigger	Action	Who does it
Act now	High hazard, narrow interval, powered equipment or active treatment	Care-team call, early refill, backup-power or transport plan	VA care team
Find out	Wide interval, missing fields	3-minute check-in call to fill the gaps; re-score same day	Care team or volunteer partner
Self-serve	Medium hazard	Verified text with cooling/clean-air site, refill link, 988-press-1	Automated via VA channels
Everyday	Low hazard	Monthly wellness plan: shaded green space, movement, VA Whole Health	Automated

The tool supports clinicians and never changes a medication, decides eligibility, or contacts a veteran outside VA-approved channels.

Data plan

Real public data describes places and conditions; a documented synthetic generator describes people. Every borrowed number is cited in the README, and every made-up one carries a `synthetic=True` flag in the schema.

Real layers (ZIP or tract, all free).

Layer	What it feeds	Source	Hackathon note
Electricity-dependent Medicare beneficiaries by ZIP, monthly	Prior on powered-equipment prevalence; outage scenario	HHS emPOWER REST service	Public ArcGIS endpoint, JSON, 2,000 records per query
Heat Vulnerability Index (1–5 by neighborhood)	ZIP-level heat prior	NYC Health HVI	Built from surface temperature, green space, home AC, income

Layer	What it feeds	Source	Hackathon note
Heat mortality facts	Severity weights, pitch numbers	NYC 2026 heat mortality report	About 500 heat-exacerbated deaths a year; indoor deaths almost always without AC running
Air quality and smoke	Daily PM2.5 covariate	EPA AirNow API	Use 2023 Canadian-smoke days as the replay scenario
Forecasts and alerts	Event trigger, lead time	NWS API (api.weather.gov)	Free, no key
Flood and hurricane risk	Long-run hazard prior	FEMA National Risk Index	Tract level
Social vulnerability, chronic disease rates	ZIP random-effect covariates	CDC SVI, CDC PLACES	Tract → ZIP crosswalk
Veteran counts by age, era, disability	Cohort weights	ACS tables S2101 / S2102	ZCTA level
Care and crisis locations	Action plans	VA Facilities API, SAMHSA locator	Cooling centers from NYC Open Data when active

Synthetic cohort. Start from the VA's public Synthea release of [10,000 synthetic veteran records](#) with 500+ clinical concepts and SDoH fields. Then:

1. Re-home each record to an NYC ZIP, sampling ZIPs in proportion to ACS veteran counts by age band, so the map is right.
2. Add fields Synthea lacks: deployment era and burn-pit dose, PTSD severity, home AC (drawn from HVI neighborhood rates), powered equipment (drawn from emPOWER counts), active cancer treatment, heat-sensitive medications (flag diuretics, anticholinergics, beta-blockers, antipsychotics from the Synthea medication table).
3. Simulate 120 days of daily outcomes from a *known* generative model with fixed true coefficients. This is the ground truth the model must recover.
4. Hide 20 percent of the AC and deployment fields at random to create real missingness.

Known gaps, stated as assumptions.

- No veteran-level data exists publicly, so every individual is synthetic and every individual-level effect is a simulation assumption.
- Eligibility and enrollment are not public; the demo assumes all 10,000 are enrolled with a care team.
- Burn-pit dose is unobserved in reality, so the model treats it as a latent variable with era-based priors, not a known number.
- Outage feeds are not archived, so the outage scenario is scripted from a past Con Ed event, not replayed from a feed.
- Intervention effects τ come from published outreach studies and are shown as adjustable priors, not measured here.

The Bayesian model

Leeward fits a hierarchical discrete-time hazard model: for each veteran i , day t and need k (breathing flare-up, heat illness, mental-health crisis, treatment or medication gap), the probability the need arises that day. Five upgrades over the base logistic model, each tied to a thing judges can see.

1. Daily hazard with distributed lags. Heat harms over 0–3 days and smoke over 0–2. Lag weights are constrained to a smooth curve (two spline coefficients per hazard), so the model learns the shape from four numbers, not forty.

$$\operatorname{logit} h_{ikt} = \alpha_k + \phi_{z(i),k} + \beta_k^{\text{top}} H_i + \gamma_k \Lambda_i + \sum_{l=0}^3 \delta_{kl} f(\left(C_{z(i)}, t-l\right) + \theta_k^{\text{top}} \left(C_{z(i)}, t\right) \otimes V_i) + \eta_k U_i + \xi_i$$

Term	Meaning	Example
$\phi_{z,k}$	ZIP effect with an ICAR spatial prior, so neighboring ZIPs share strength	East Harlem borrows from Central Harlem, not from Staten Island
H_i	Health status	Age, COPD, CHF, diabetes, active cancer treatment, heat-sensitive meds
Λ_i	Latent cumulative exposure dose (see 2)	Burn-pit years $\times$ era, trauma, years of local heat
$f(C)$	Hazard intensity, nonlinear	Piecewise-linear heat index with a hinge at the NYC 82°F threshold; PM2.5

Term	Meaning	Example
$C \otimes V$	Condition $\times$ vulnerability interactions	Smoke day $\times$ COPD; heat $\times$ diuretic; outage $\times$ oxygen concentrator
U_i	Past care use	ER visits, missed refills, missed appointments
ξ_i	Shared frailty across the four needs	A veteran prone to one need is prone to others; captures mind-body coupling from the WTC findings

2. Latent exposure with measurement error. Deployment exposure is never observed exactly. Λ_i is a latent variable with an era-based prior (Gulf War and post-9/11 eras get higher mean dose) and a noisy indicator when a record contains a PACT Act presumptive diagnosis. Wide uncertainty in Λ flows into wide uncertainty in h , which is what routes that veteran to the *Find out* tier.

3. Weakly informative priors from the literature, shown, not hidden. Priors for trauma and airborne-exposure effects are centered on WTC Registry and PACT Act evidence with wide scales. The demo has a slider that halves or doubles the prior scale and re-scores live; rankings barely move, which is the point.

4. Fit on sufficient statistics, not rows. 10,000 veterans $\times$ 120 days $\times$ 4 needs is 4.8 million Bernoulli rows. Grouping by ZIP $\times$ covariate stratum $\times$ day gives binomial cells with identical likelihood and about 50 $\times$ fewer rows. NumPyro NUTS on JAX fits this in minutes on a laptop; the posterior is cached, so live scoring of a new forecast is a matrix multiply.

5. Uncertainty decomposition. For each veteran the posterior predictive is split into aleatoric (irreducible daily chance) and epistemic (parameter and latent-variable uncertainty). Epistemic width is what the decision layer treats as value of information.

Priors, stated.

Parameter	Prior	Source of center
Heat-by-diuretic interaction θ	Normal(0.4, 0.5) on log-odds	Heat-medication literature; wide
Smoke-by-COPD interaction θ	Normal(0.6, 0.5)	PACT Act presumptive list; wide

Parameter	Prior	Source of center
PTSD main effect β	Normal(0.3, 0.3)	WTC Registry mind-body findings
Lag curve δ	Random walk, $\sigma = 0.2$	Smoothness only
ZIP effect ϕ	ICAR, $\tau \sim \text{Half-Normal}(1)$	Spatial smoothness only
Frailty ξ	Normal(0, σ_ξ), $\sigma_\xi \sim \text{Half-Normal}(0.5)$	Weak

Outputs. Per veteran per day: posterior mean and 80 percent interval for each need, epistemic share, and the top three drivers (from posterior contribution decomposition, not SHAP on a black box). Per ZIP per day: expected count of each need with an interval, for staffing. Per care team: today's action list from the decision layer.

Why not XGBoost. A tree model would fit the synthetic data as well or better. It cannot carry a latent exposure dose, cannot say how much of a prediction is ignorance, cannot share strength across small ZIPs, and cannot update from 30 logged outcomes without a full retrain. The pitch names this trade openly.

Validation

Because the cohort comes from a generator with known coefficients, Leeward can prove it recovers the truth, which no team using hand-labeled synthetic profiles can do. Five checks, all rendered as one "Model report" screen in the demo.

Check	What it shows	Pass bar
Parameter recovery	True coefficient inside the 90 percent posterior interval	90 percent of parameters, plotted as dot-and-whisker against truth
Calibration	Predicted 30 percent risks happen about 30 percent of the time, per need, per week	Expected calibration error under 0.03; reliability curve on screen
Held-out discrimination	Days 91–120 unseen during fit	PR-AUC per need against a no-climate baseline; report both

Check	What it shows	Pass bar
Posterior predictive check	Simulated daily counts by ZIP against observed counts	Observed inside the 90 percent band on 90 percent of days
Simulation-based calibration	50 small refits on fresh draws from the prior	Rank histograms roughly flat; run overnight Saturday

Decision quality, the number to say on stage. Fix a daily capacity of K calls. Compare harm averted, in simulated need-days prevented, when ranking by Leeward’s expected harm averted against three baselines: rank by age, rank by a CAN-like chronic-condition count, and random. Report the ratio. This single bar chart is the Impact slide.

Ablations (one table, generated by the pipeline): no climate terms, no lags, no interactions, no latent exposure, no spatial prior. Each row reports calibration and harm averted at K, so judges see what every model component buys.

Fairness audit. Calibration and false-negative rate by borough, by neighborhood HVI band, and by the race and ethnicity fields Synthea carries. The audit is a pass only if no group’s false-negative rate exceeds the cohort rate by more than 20 percent relative; if it does, the demo shows the gap rather than hiding it, and the decision layer can apply a group-level capacity floor.

What we will not claim. Nothing here validates the model on real veterans. It validates that the machinery is correct and the uncertainty is honest, which is the precondition for a VA pilot.

From prediction to action

Every action Leeward proposes already exists inside the VA or a partner; the tool routes to it earlier and through a channel the veteran can trust. This is the section that covers opportunity areas 04 and 05, which the base proposal skipped.

Action catalog, keyed to need and lead time.

Need	5 days out	2 days out	During	Existing program used
Medication gap	Early refill via My HealtheVet	Confirm delivery or local pickup	Emergency Pharmacy Refill Program voucher at retail pharmacies when activated	VA disaster help ; Meds by Mail local refill for CHAMPVA
Heat illness	Check AC; HEAP cooling application if none	Cooling-center match with step-free transit route	Wellness call on day 2 of the wave	NYC cooling centers, HEAP
Breathing flare-up	Inhaler and oxygen supply check	Clean-air room match; keep windows closed advisory	Same-day tele-visit slot	VA mobile pharmacy and oxygen resupply when deployed
Treatment gap (chemo, dialysis, oxygen)	Backup-power plan; register with utility medical-needs list	Confirm ride or reschedule infusion	Care-team call within 2 hours of outage	HHS emPOWER planning, VA transport
Mental-health crisis	Move therapy to phone if outage likely	Buddy check via peer partner	Any crisis signal → Veterans Crisis Line, 988 press 1	VCL, VA Whole Health

Verified outreach, the anti-exploitation piece (05). Scammers pose as FEMA staff, adjusters and charities after disasters, and VA's own fraud line, VSAFE at 833-388-7233, exists for this ([VA News, Aug 2026](#)). Leeward's messages are built so a veteran can tell them apart from a scam:

- Every message goes only through VA channels the veteran already uses (VEText, My HealtheVet secure message, care-team phone). Leeward never sends from a new number.

- Every message carries a 4-word verification phrase the veteran can read back to the care team, and a line that says *the VA will never ask you to pay, wire money or share your bank details*.
- Every message includes the VSAFE number and the Veterans Crisis Line.
- A one-page "After the storm" scam card ships with the high-tier plan, generated from the VA and FTC guidance, not written by us.

Coordination (04). The care team's action list exports, with consent flags, as a partner sheet: which veterans opted in to a Team Rubicon wellness check, which need a Combined Arms ride, which need a HUD-VASH housing contact. One list, one owner (the VA care team), many hands. The demo shows the export and a mock partner acknowledgment closing the loop.

Outcome log. Each action records done, not done, reached, not reached, and whether the need occurred. These 20–30 rows a week are exactly what the Bayesian model can absorb without a retrain, which is how the WTC-derived priors get tested rather than trusted.

Demo storyline and pitch

Most teams will open with a map. Leeward opens with one veteran, because Impact is the biggest rubric line and a person carries impact better than a heatmap. Total time 3 minutes talking, 2 minutes live demo, with the demo pre-recorded as a fallback.

The story. "Walter" (synthetic): 71, Bronx, Gulf War era, COPD on the PACT Act presumptive list, oxygen concentrator at night, takes a diuretic, no AC on record. It is Monday. NWS forecasts a coastal storm Wednesday with outages likely, then a three-day heat wave from Friday. Today Walter is one of 1,200 patients on his care team's panel.

Scene sequence (2 minutes, live).

1. Forecast panel: NWS alert and AirNow feed for NYC; the model re-scores 10,000 veterans in under a second.
2. Map: expected needs by ZIP for the next 7 days, with intervals; the Bronx and East Harlem light up. One click on a ZIP shows why (HVI 5, low AC, high emPOWER count).
3. Care-team screen: today's action list, cut at 40 calls. Walter is number 3. His card shows 62 percent (45–78) chance of a treatment gap by Thursday, top drivers: oxygen concentrator × outage, diuretic × heat, no AC. Action: call today, backup-power plan, early refill, cooling-center ride Friday.
4. Capacity slider: drag from 40 to 20 calls; the list re-ranks and the harm-averted counter drops; drag to 80 and the *Find out* tier fills with wide-interval veterans.

5. The message Walter gets: VA channel, verification phrase, 988 press 1, VSAFE number, the never-pay line.
6. Model report: recovery plot, calibration curve, harm averted versus the age-ranked baseline, fairness table by borough.

Pitch script (3 minutes).

- 0:00 The problem in one number: about 500 New Yorkers die of heat-exacerbated causes each year, nearly all indoors without AC running ([NYC Health](#)). Veterans carry every heat risk factor plus two the city does not count: deployment exposure and trauma.
- 0:30 Walter. Who he is, what Monday looks like, what usually happens: nobody calls until Friday's ER visit.
- 1:00 What Leeward does: three questions, who, when, what changes if we act. Show the flow diagram once.
- 1:30 Why Bayesian, in plain words: it starts from what the WTC Registry and the PACT Act already taught us, it says how sure it is, and uncertainty decides who gets a 3-minute call versus a refill and a ride.
- 2:00 Proof: the model recovers the truth we planted, it is calibrated, and at 40 calls a day it averts $N\times$ the harm of calling the oldest patients first.
- 2:30 Areas 04 and 05: one list, many partners; messages a scammer cannot imitate.
- 2:50 Ask: a one-summer pilot with one VA medical center's care team, priors validated on de-identified data under IRB.

Presentation rules the team agrees to.

- One person talks; one person drives; never the same person.
- Every slide has one number or one face, never a bullet list.
- Say "synthetic" out loud three times: at the start, at the model report, and at the ask.
- Name the limit before the judges do: WTC priors are assumptions; the demo shows they do not drive the ranking.
- Rehearse the capacity-slider moment; it is the part people remember.

Submission pack. Two-page PDF (this doc, trimmed), 90-second video of the live flow, public GitHub repo with a one-command `make demo`, and a slide deck of 8 slides.

Claude Code build plan

The build is four independent tracks that meet at a single `data/posterior.nc` file and a single `scores.parquet`. That contract lets four Claude Code sessions (or four people) run in parallel from hour one. Total budget: about 26 working hours across Saturday afternoon and Sunday morning.

Repo layout.

```

leeward/
  CLAUDE.md                # project brief, contracts, do-not-do list
  Makefile                 # make data | fit | score | demo | report
  pyproject.toml           # python 3.11, numpyro, jax, pandas, polars,
fastapi
  data/
    raw/                   # synthea csvs, emPOWER json, HVI, ACS
(gitignored, fetched by script)
  cohort.parquet           # synthetic NYC cohort, generator version stamped
  truth.json               # planted coefficients
  posterior.nc              # cached ArviZ InferenceData
  scores.parquet           # veteran × day × need posterior summaries
  leeward/
    ingest/                # empower.py, hvi.py, acs.py, nws.py, airnow.py
(each: fetch → parquet)
  cohort/                  # rehome.py, augment.py, simulate.py (generator
with truth.json)
  model/                   # hazard.py (numpyro), priors.py, fit.py,
score.py, decompose.py
  decision/                # eha.py, allocate.py (greedy knapsack), tiers.py
  outreach/                # messages.py (templates, verification phrase),
export.py
  eval/                    # recovery.py, calibration.py, ppc.py, sbc.py,
fairness.py, ablate.py
  api/                     # fastapi: /forecast, /scores, /actions, /report
  ui/                      # single-file React or Next app: map, care-team
list, slider, report
  scenarios/               # storm_then_heat.yaml, smoke_2023.yaml
  tests/                   # shape tests, one-veteran smoke test,
calibration guard
  docs/                    # two-pager, slides, sources.md

```

CLAUDE.md, the whole file.

Leeward — veteran care continuity under climate events

What this is

Bayesian daily-hazard model + capacity-aware decision layer for VA care teams.

Hackathon demo. Synthetic data only. Never write code that could ingest real PHI.

Contracts (do not change without updating all consumers)

- data/cohort.parquet: one row per veteran; columns in leeward/cohort/schema.py; every simulated column has a sibling ``<col>_synthetic=True`.`
- data/scores.parquet: columns [veteran_id, date, need, p_mean, p_lo80, p_hi80, p_epistemic_share, driver_1, driver_2, driver_3].
- data/posterior.nc: ArviZ InferenceData; var names match leeward/model/priors.py.
- API responses are typed with pydantic models in leeward/api/schemas.py.

Stack

Python 3.11, NumPyro + JAX (CPU ok), polars for ETL, FastAPI, React + deck.gl for the map.

Use `make fit` (~8 min CPU). Never run inference inside an API request; score from cache.

Rules

- Cite every real number in docs/sources.md with a URL. Synthetic numbers say so.
- No SHAP; drivers come from posterior linear-predictor contributions.
- Fairness audit must run in `make report`; a failing audit is shown, not suppressed.
- Messages must include: VA channel only, verification phrase, 988 press 1, VSAFE 833-388-7233, never-pay line.
- Tests before merge: `pytest -q` must pass; `make demo` must boot in < 60 s from a clean clone.

Team map

Track A data/cohort · Track B model/eval · Track C decision/api · Track D ui/docs

Phases and hours.

Hour	Track A: data + cohort	Track B: model + eval	Track C: decision + API	Track D: UI + pitch
0–2	Fetch Synthea VA release, emPOWER REST, HVI, ACS; write <code>schema.py</code>	Write <code>hazard.py</code> on a 200-veteran toy cohort; NUTS runs end to end	Stub API with fake scores matching the contract	Skeleton app on stub API; map renders NYC ZIPs
2–6	<code>rehome.py</code> and <code>augment.py</code> ; cohort of 10,000 with planted truth	Sufficient-statistic binning; fit on full cohort; <code>posterior.nc</code> cached	<code>eha.py</code> , <code>allocate.py</code> , <code>tiers.py</code> on stub scores	Care-team list, veteran card, capacity slider
6–10	<code>simulate.py</code> 120 days; two scenario YAMLS; 20 percent missingness	<code>score.py</code> , <code>decompose.py</code> (drivers, epistemic share)	Wire real scores; <code>/actions</code> endpoint; export sheet	Message preview screen; scam card; forecast panel
10–16 (overnight)	Sources doc; data README	<code>recovery.py</code> , <code>calibration.py</code> , <code>ppc.py</code> , <code>ablate.py</code> ; start SBC job	Baselines (age, chronic-count, random); harm-averted chart data	Model report page from eval outputs; slides v1
16–22	Freeze cohort; tag release	Fairness audit; prior-sensitivity slider endpoint	Verification-phrase generator; outcome log endpoint	Rehearse; record 90-second video; two-pager
22–26	Clean-clone <code>make demo</code> test	Final numbers into slides	Bug bash	Final rehearsal with the capacity-slider beat

How to drive Claude Code on each track. Start every session with the same three lines: read `CLAUDE.md` , read the track’s module, run `make test` . Then give one task per prompt with the acceptance test written in the prompt, for example: "Implement `allocate.py` : greedy selection of actions maximizing summed EHA under `capacity: dict[action, int]` ; add a test with 5 veterans where the answer is hand-checkable." Ask for a plan before any change touching a contract file. Use a scratch branch per track and merge through the Makefile targets, never by hand.

Cut list if time runs short, in order. SBC → spatial ICAR prior (fall back to independent ZIP effects) → prior-sensitivity slider → partner export → deck.gl map (fall back to a static Plotly choropleth). The decision layer, calibration plot and Walter’s card are never cut.

Risks, limits and next steps

The biggest hackathon risk is not the model; it is a fit that has not finished by Sunday morning. The mitigation is baked in: fit on binomial cells, cache the posterior Saturday night, and never run inference in the demo path.

Risk	Likelihood	Mitigation
NUTS divergences on the latent-exposure block	Medium	Non-centered parameterization; if it persists, fix Λ at its prior mean for the demo and say so
emPOWER or NWS endpoint down during demo	Low	Snapshot every feed to <code>data/raw</code> on Saturday; demo runs offline
Synthea VA release lacks a needed field	Medium	Augment step already owns those fields; nothing depends on Synthea beyond demographics, conditions and medications
Judges read "Bayesian" as slow or academic	Medium	Lead with Walter and the capacity slider; the model report comes last
Fairness audit fails	Possible	Show it; explain the capacity floor; it reads as rigor, not failure

Limits stated on stage and in the two-pager.

- Every veteran is synthetic; every individual-level effect is an assumption we planted and then recovered.

- WTC Registry and PACT Act evidence set prior centers only; the ranking is shown to be robust to doubling or halving them.
- Intervention effects τ are literature priors, adjustable in the UI, not measured here.
- The tool proposes; clinicians approve every high-tier action; nothing changes a medication or a benefit.

Next steps after the hackathon.

1. Validate priors on de-identified VA data with a VA research partner under IRB, replacing planted truth with real outcomes.
2. Pilot for one summer with one NYC VA medical center care team; success metric is harm averted at fixed staff time, measured against the prior summer.
3. Connect the everyday tier to VA Whole Health and NYC Parks shade data so the same model serves veterans on ordinary days, not only during events.
4. Publish the cohort generator and evaluation harness as open source so other teams can benchmark against a shared synthetic NYC cohort.

Sources

- [Health in Climate 2025 Devpost rubric](#)
- [VA Synthea synthetic veteran dataset, 10,000 records](#)
- [HHS emPOWER public REST service](#)
- [NYC Heat Vulnerability Index](#)
- [NYC 2026 heat-related mortality report](#)
- [VA disaster help and fraud line](#)
- [VA News, natural-disaster fraud prevention, Aug 2026](#)
- [Hackathon problem page](#)